H9: Electromagnetic waves

9.2 electromagnetic waves in a vacuum

9.2.1 the wave equation for E and B

wave equation for E and B

In regions of space with no charge or current, we have:

$$\text{(i)}\quad \nabla\cdot\mathbf{E}=0,\quad \text{(iii)}\quad \nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t},$$

$$\text{(ii)}\quad \nabla\cdot\mathbf{B}=0,\quad \text{(iv)}\quad \nabla\times\mathbf{B}=\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}.$$

We can decouple E and B by taking:

$$\nabla\times(\nabla\times\mathbf{E})=\nabla(\nabla\cdot\mathbf{E})-\nabla^2\mathbf{E}=\nabla\times\left(-\frac{\partial\mathbf{B}}{\partial t}\right)$$

$$=-\frac{\partial}{\partial t}(\nabla\times\mathbf{B})=-\mu_0\epsilon_0\frac{\partial^2\mathbf{E}}{\partial t^2},$$

and:

$$\nabla\times(\nabla\times\mathbf{B})=\nabla(\nabla\cdot\mathbf{B})-\nabla^2\mathbf{B}=\nabla\times\left(\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}\right)$$

$$=\mu_0\epsilon_0\frac{\partial}{\partial t}(\nabla\times\mathbf{E})=-\mu_0\epsilon_0\frac{\partial^2\mathbf{B}}{\partial t^2}.$$

then we find:

$$\nabla^2\mathbf{E}=\mu_0\epsilon_0\frac{\partial^2\mathbf{E}}{\partial t^2},\quad \nabla^2\mathbf{B}=\mu_0\epsilon_0\frac{\partial^2\mathbf{B}}{\partial t^2}.$$

which are two wave equations of waves travelling at the speed:

$$v=\frac{1}{\sqrt{\epsilon_0\mu_0}}$$

9.2.2 monochromatic plane waves

plane waves in E and B

suppose we have two plane waves travelling in z-direction:

$$\tilde{\mathbf{E}}(z,t)=\tilde{\mathbf{E}}_0e^{i(kz-\omega t)},\quad \tilde{\mathbf{B}}(z,t)=\tilde{\mathbf{B}}_0e^{i(kz-\omega t)},$$

with E_0 and B_0 the amplitude and $\omega=kc$ the angular frequency

> now Maxwell's equations put constraints on these waves:

<u>∇E=0 and ∇B=0:</u> $(\tilde{E}_0)_z=(\tilde{B}_0)_z=0.$

>> the E and B are transverse waves,
ie: their amplitudes are perpendicular to the direction of propagation

<u>∇xE=-∂B/∂t</u>: $-k(\tilde{E}_0)_y=\omega(\tilde{B}_0)_x,\quad k(\tilde{E}_0)_x=\omega(\tilde{B}_0)_y,$

or thus:

$$\tilde{\mathbf{B}}_0=\frac{k}{\omega}(\hat{\mathbf{z}}\times\tilde{\mathbf{E}}_0).$$

>> E and B are mutually perpendicular
> they are in phase and their amplitudes are related by:

$$B_0=\frac{k}{\omega}E_0=\frac{1}{c}E_0.$$

∇xB=$\mu_0\epsilon_0$∂E/∂t doesn't imply any new constraints

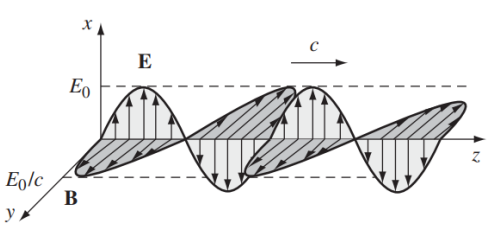
>> we conclude that E and B are transverse waves, in phase, that are perpend. to each other
thus:

$$\tilde{\mathbf{E}}(\mathbf{r},t)=\tilde{E}_0e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}\,\hat{\mathbf{n}},$$

$$\tilde{\mathbf{B}}(\mathbf{r},t)=\frac{1}{c}\tilde{E}_0e^{i(\mathbf{k}\cdot\mathbf{r}-\omega t)}(\hat{\mathbf{k}}\times\hat{\mathbf{n}})=\frac{1}{c}\hat{\mathbf{k}}\times\tilde{\mathbf{E}},$$

with **k** the vector of magnitude k pointing in the direction of propagation
n the polarization vector

$$\hat{\mathbf{n}}\cdot\hat{\mathbf{k}}=0.$$



9.2.3 energy and momentum in electromagnetic waves	
u and **S** for E and B plane waves	If we have plane waves for E and B, we also know: $u = \frac{1}{2}\left(\epsilon_0 E^2 + \frac{1}{\mu_0}B^2\right)$ with $B^2 = \frac{1}{c^2}E^2 = \mu_0\epsilon_0 E^2,$ then: $u = \epsilon_0 E^2 = \epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta).$ >> the E and B fields have equal contribution to the energy per unit volume We also have for $\mathbf{S} = \frac{1}{\mu_0}(\mathbf{E} \times \mathbf{B}).$ that: $\mathbf{S} = c\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)\,\hat{\mathbf{z}} = cu\,\hat{\mathbf{z}}.$
momentum of E and B plane waves	The momentum density in the E and B fields is given by: $\mathbf{g} = \frac{1}{c^2}\mathbf{S}.$ then we find: $\mathbf{g} = \frac{1}{c}\epsilon_0 E_0^2 \cos^2(kz - \omega t + \delta)\,\hat{\mathbf{z}} = \frac{1}{c}u\,\hat{\mathbf{z}}.$
expectation values of u,S,g	if we measure these properties of the lightwaves, the fluxtuation of the waves is so rapid that we measure the average value of these quantities: $\langle u\rangle = \frac{1}{2}\epsilon_0 E_0^2, \quad \langle \mathbf{S}\rangle = \frac{1}{2}c\epsilon_0 E_0^2\,\hat{\mathbf{z}}, \quad \langle \mathbf{g}\rangle = \frac{1}{2c}\epsilon_0 E_0^2\,\hat{\mathbf{z}}.$
intensity of the wave	this is the average power per unit area transported by the wave $I \equiv \langle S\rangle = \frac{1}{2}c\epsilon_0 E_0^2.$
radiation pressure	in a time Δt the momentum transfer is: $\Delta\mathbf{p} = \langle \mathbf{g}\rangle Ac\,\Delta t$ now define: $P = \frac{1}{A}\frac{\Delta p}{\Delta t} = \frac{1}{2}\epsilon_0 E_0^2 = \frac{I}{c}.$
9.3 electromagnetic waves in matter	
9.3.1 propagation in linear media	
Maxwell eqn in linear media	inside matter with no free charge or free current: $\left.\begin{array}{ll}\text{(i)} \quad \nabla\cdot\mathbf{D} = 0, & \text{(iii)} \quad \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t},\\ \text{(ii)} \quad \nabla\cdot\mathbf{B} = 0, & \text{(iv)} \quad \nabla\times\mathbf{H} = \frac{\partial\mathbf{D}}{\partial t}.\end{array}\right\}$ now for linear media: $\mathbf{D} = \epsilon\mathbf{E}, \quad \mathbf{H} = \frac{1}{\mu}\mathbf{B},$ thus: $\left.\begin{array}{ll}\text{(i)} \quad \nabla\cdot\mathbf{E} = 0, & \text{(iii)} \quad \nabla\times\mathbf{E} = -\frac{\partial\mathbf{B}}{\partial t},\\ \text{(ii)} \quad \nabla\cdot\mathbf{B} = 0, & \text{(iv)} \quad \nabla\times\mathbf{B} = \mu\epsilon\frac{\partial\mathbf{E}}{\partial t},\end{array}\right\}$ which resembles the eqn in a vacuum, but $\mu_0\varepsilon_0$ is now με > em waves propagate through linear homogeneous media at a speed $v = \frac{1}{\sqrt{\epsilon\mu}} = \frac{c}{n},$ with n the index of refraction: $n \equiv \sqrt{\frac{\epsilon\mu}{\epsilon_0\mu_0}}$
energy density, Poynting vector and intensity	if we just replace μ_0>μ and ε_0>ε then: $u = \frac{1}{2}\left(\epsilon E^2 + \frac{1}{\mu}B^2\right) \qquad \mathbf{S} = \frac{1}{\mu}(\mathbf{E}\times\mathbf{B}). \qquad I = \frac{1}{2}\epsilon v E_0^2.$
boundary conditions	when a lightwave goes from one medium to another, it must respect: $\left.\begin{array}{ll}\text{(i)} \quad \epsilon_1 E_1^{\perp} = \epsilon_2 E_2^{\perp}, & \text{(iii)} \quad \mathbf{E}_1^{\parallel} = \mathbf{E}_2^{\parallel},\\ \text{(ii)} \quad B_1^{\perp} = B_2^{\perp}, & \text{(iv)} \quad \frac{1}{\mu_1}\mathbf{B}_1^{\parallel} = \frac{1}{\mu_2}\mathbf{B}_2^{\parallel}.\end{array}\right\}$